Power BI Project: Store Sales & Inventory Dashboard

Intermediate level · Built on the data from your SQL inventory project · About 12–15 hours over 1–2 weeks

Why this project

You already have a year of store data (32,000 bills, 90,000+ sale lines) in SQLite. Turning it into a Power BI dashboard shows the full chain recruiters look for in a data analyst: **SQL → cleaning → data model → DAX → dashboard → insight**. It also links your two resume projects into one story, which makes for a strong interview answer.

You will build	A 3-page interactive report: **Overview**, **Products & Suppliers**, **Stock Alerts**.
Skills shown	Power Query cleaning, star-schema modelling, a date table, DAX with CALCULATE and time intelligence, drill-through, slicers, conditional formatting, tooltips, bookmarks.
You need	Power BI Desktop (free, Windows), your SQLite database, and an hour to read through this guide first.

Step 1 – Export your data from SQLite

CSV files are the simplest way in. In a terminal, open your database and export each table you need. Your table and column names may differ; adjust them to match your schema.

```
sqlite3 store.db
.headers on
.mode csv
.output sales_lines.csv
SELECT * FROM sale_lines;
.output bills.csv
SELECT * FROM bills;
.output products.csv
SELECT * FROM products;
.output suppliers.csv
SELECT * FROM suppliers;
.output purchases.csv
SELECT * FROM supplier_purchases;
.output stdout
```

Tip: you can also export one of your existing report queries (top sellers, P&L) as a CSV to cross-check your DAX numbers later. If both match, you know your model is right. Mention this in interviews.

Step 2 – Clean the data in Power Query

- **Home → Get Data → Text/CSV** for each file, then click **Transform Data**.
- Set correct data types: dates as Date, prices as Fixed Decimal, IDs as Whole Number or Text.
- Trim and clean text columns (product names, categories) so "Rice " and "rice" don't count as two items.
- Remove duplicate rows and blank rows; check for nulls in price and quantity.
- If sale lines don't carry the date, **Merge Queries** with bills on the bill ID to bring the date across.
- Add a column **Line Revenue = Quantity × Unit Price** and **Line Cost = Quantity × Cost Price** if you store cost.
- Rename tables: **FactSales**, **FactPurchases**, **DimProduct**, **DimSupplier**.

Step 3 – Build a star schema

In **Model view**, keep facts in the middle and dimensions around them. Use one-to-many relationships with single-direction filtering (dimension → fact).

```
DimDate[Date]            1 --> *  FactSales[SaleDate]
DimProduct[ProductID]    1 --> *  FactSales[ProductID]
DimDate[Date]            1 --> *  FactPurchases[PurchaseDate]
DimProduct[ProductID]    1 --> *  FactPurchases[ProductID]
DimSupplier[SupplierID]  1 --> *  FactPurchases[SupplierID]
```

Create the date table with DAX (**Modeling → New Table**), then **Mark as Date Table**:

```
DimDate =
ADDCOLUMNS (
    CALENDAR ( MIN ( FactSales[SaleDate] ), MAX ( FactSales[SaleDate] ) ),
```

```
    "Year", YEAR ( [Date] ),
    "Month No", MONTH ( [Date] ),
    "Month", FORMAT ( [Date], "MMM yyyy" ),
    "Weekday", FORMAT ( [Date], "ddd" )
)
```

Sort the **Month** column by a year-month number so months appear in order, not alphabetically.

Step 4 – Write the DAX measures

Put all measures in one empty table called **_Measures** to keep the model tidy.

```
Total Revenue = SUM ( FactSales[LineRevenue] )
Total Cost    = SUM ( FactSales[LineCost] )
Gross Margin  = [Total Revenue] - [Total Cost]
Gross Margin % = DIVIDE ( [Gross Margin], [Total Revenue] )
Bills Count   = DISTINCTCOUNT ( FactSales[BillID] )
Avg Bill Value = DIVIDE ( [Total Revenue], [Bills Count] )

Revenue Prev Month =
    CALCULATE ( [Total Revenue], DATEADD ( DimDate[Date], -1, MONTH ) )

Revenue MoM % =
    VAR Prev = [Revenue Prev Month]
    RETURN IF ( NOT ISBLANK ( Prev ), DIVIDE ( [Total Revenue] - Prev, Prev ) )

Running Revenue =
    CALCULATE (
        [Total Revenue],
        FILTER ( ALL ( DimDate[Date] ), DimDate[Date] <= MAX ( DimDate[Date] ) )
    )

Revenue Share % =
    DIVIDE ( [Total Revenue], CALCULATE ( [Total Revenue], ALL ( DimProduct ) ) )

Low Stock Items =
    COUNTROWS (
        FILTER ( DimProduct, DimProduct[StockQty] < DimProduct[ReorderLevel] )
    )
```

Check each measure against your SQL query results before building visuals. Understand what **CALCULATE**, **ALL** and **filter context** do in each measure. These are the most common Power BI interview questions.

Step 5 – Build the three pages

Page 1 · Overview

- KPI cards: Total Revenue, Gross Margin %, Bills Count, Avg Bill Value.
- Line chart: Total Revenue and Running Revenue by month. Add Revenue MoM % to the tooltip.
- Bar chart: revenue by weekday, which shows the store's busiest days.
- Slicers: month range and product category, synced across all pages (**View → Sync slicers**).

Page 2 · Products & Suppliers

- Top 10 products by revenue (bar chart with **Top N** filter) and a matrix: category → product, with revenue, margin % and share %.
- Conditional formatting on margin %: red below 10%, green above 25% (pick thresholds that fit your data).
- Supplier table: purchase value, number of orders and average delay in days, if your data has expected vs actual dates.
- **Drill-through:** a hidden detail page where right-clicking a product shows its monthly sales and stock history.

Page 3 · Stock Alerts

- Card: Low Stock Items. Table: products below reorder level, sorted by how fast they sell.
- Link it to your SQL reorder rule: show which items your rule would reorder this week.
- Add a **bookmark** button that switches between "All items" and "Only alerts".

Step 6 – Polish

- Use one colour theme (**View → Themes**) and at most 3–4 colours.

- Write titles that state the insight, e.g. "Saturdays bring 22% of weekly revenue", using your real numbers.
- Align visuals to a grid, give every chart axis titles, and remove unneeded gridlines.
- Add a small text box on each page explaining what it answers. Recruiters skim.

Step 7 – Export and share the project

Save the file	**File → Save As** → **Store_Sales_Dashboard.pbix**. This is the full project, including data and model.
PDF copy	In Power BI Desktop: **File → Export → Export to PDF**. This saves every page as a static PDF that anyone can open.
Screenshots & demo	Take a screenshot of each page, and record a 30–60 second screen video of the slicers and drill-through (Windows: **Win + Alt + R** with Xbox Game Bar, or the Snipping Tool's video mode).
GitHub repo	Create a repo with the .pbix, the PDF, the screenshots, your SQL export queries and a README covering the problem, data, model diagram, key measures and 3–4 insights. Put this link on your resume and LinkedIn.
Power BI Service (optional)	Publishing online needs a work or school account, so try signing in with your NSUT email. **Publish to web** is often switched off by college admins. If it is, the GitHub repo plus PDF is enough for applications.
LinkedIn	Post 2–3 screenshots with a short write-up of what you found, and add the repo to your **Featured** section.

Step 8 – Be ready for these interview questions

- What is the difference between a measure and a calculated column? Why did you use measures here?
- What does CALCULATE do? How does ALL change the result of Revenue Share %?
- Why a star schema? What problems do bidirectional relationships cause?
- Why does Revenue MoM % show blank for the first month?
- How did you check that your dashboard numbers were correct? (Answer: matched them to your SQL reports.)